P R O J E C T

Heart Disease Prediction

A Comprehensive Technical Report

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Declaration

I hereby declare that the project titled **“Heart Disease Prediction using Apache Spark”** is my original work carried out as part of my academic curriculum. This project has not been submitted previously for any degree or diploma. All sources of information used have been properly acknowledged.

Abstract

Heart disease is one of the leading causes of mortality worldwide. Early prediction can significantly reduce risks and improve patient outcomes. This project focuses on predicting heart disease using machine learning techniques implemented with Apache Spark for efficient big data processing. The system analyzes medical attributes such as age, cholesterol level, blood pressure, and heart rate to determine the likelihood of heart disease. Using Spark MLlib, scalable models are trained and evaluated to provide accurate predictions. The project demonstrates how distributed computing enhances performance and handles large healthcare datasets effectively.

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Chapter 1: Introduction

Heart disease remains a major global health concern, causing millions of deaths each year. Traditional diagnostic methods can be time-consuming and may not always provide early detection. With the advancement of data analytics and machine learning, predictive models can assist healthcare professionals in identifying high-risk patients.
This project aims to develop a heart disease prediction system using Apache Spark. Spark provides a distributed computing environment, making it suitable for handling large-scale healthcare datasets. By leveraging machine learning algorithms, the system can analyze patterns and provide accurate predictions, helping in early diagnosis and prevention.

Chapter 2: Literature Review and Problem Statement

Literature Review
Previous studies have explored various machine learning algorithms such as Logistic Regression, Decision Trees, and Random Forest for heart disease prediction. Many systems achieve good accuracy but face challenges in scalability and processing speed when handling large datasets.
Apache Spark has emerged as a powerful tool for big data analytics, offering faster computation compared to traditional methods like Hadoop MapReduce. Research shows that Spark MLlib enables efficient implementation of machine learning models with improved performance.
Problem Statement
The main problem is to develop a scalable and efficient system that can predict heart disease accurately using large datasets. Traditional systems lack speed and scalability, making it difficult to process real-time healthcare data. This project aims to overcome these limitations using Apache Spark.

Chapter 3: Methodology

The methodology involves several steps:

1. **Data Collection**
   The dataset is collected from publicly available healthcare sources containing patient medical records.
2. **Data Preprocessing**
   - **Handling missing values**
   - **Normalizing data**
   - **Feature selection**
3. **Model Selection**
   Machine learning algorithms such as Logistic Regression and Decision Trees are used.
4. **Training and Testing**
   The dataset is split into training and testing sets to evaluate model performance.
5. **Evaluation Metrics**
   Accuracy, Precision, Recall, and F1-score are used to measure performance.

Chapter 4: Implementation

The project is implemented using Apache Spark and its MLlib library.

- **Environment: PySpark / Spark MLlib**
- **Steps:**
  1. **Load dataset into Spark DataFrame**
  2. **Perform data preprocessing**
  3. **Apply feature engineering**
  4. **Train machine learning models**
  5. **Evaluate model performance**

The distributed nature of Spark ensures faster computation and efficient handling of large datasets

Chapter 5: Results and Discussion

The models were evaluated based on performance metrics:

- **Logistic Regression showed high accuracy and stability**
- **Decision Tree provided good interpretability**
- **Spark significantly reduced processing time**

The results indicate that machine learning models can effectively predict heart disease with high accuracy. The use of Spark improves scalability and performance compared to traditional systems.

Chapter 6: Conclusion and Future Work

Conclusion

This project successfully demonstrates the use of Apache Spark for heart disease prediction. The system provides accurate results and handles large datasets efficiently. It can assist healthcare professionals in early diagnosis and decision-making.

Future Work

- **Integration with real-time hospital data**
- **Use of advanced models like Deep Learning**
- **Deployment as a web or mobile application**
- **Improving accuracy with larger datasets**

Code Repository

The complete source code, dataset, dashboard application, and supporting artefacts for this project are publicly available in the following GitHub repository:

GitHub Repository

tusharverma17com-max/Heart-Disease-Prediction *Repository: diabetes-risk-prediction | Branch: main*